Group Assignment (Group of 4)

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INTRODUCTION:

The assignment focuses on 32-bit IA-32 instructions and excludes instruction prefixes. The generated machine-code representation follows the Little-Endian byte ordering used by the x86 architecture.

The instruction information is organized in an alphabetical lookup structure. Instructions such as ADD, AND, CMP, DEC, DIV, INC, MOV, MUL, SUB, and XOR are represented with their different encoding forms. The specification also defines the fields required by each instruction, including the number and type of operands, ModR/M fields, register encoding, +rd encoding, SIB requirements, immediate values, and displacement values.

This documentation also explains the role of the Two-Pass Assembler. In the first pass, the assembler identifies labels, symbols, sections, and instruction addresses. In the second pass, it uses the collected information to generate the final machine-code representation.

General IA-32 Instruction Format:

Opcode 1-2 bytes Mandatory	Mod r/m 0-1 byte optional	SIB 0-1 byte Optional	Displacement 0/1/4 bytes	Immediate 0/1/4 bytes

Description:

Field	Size	Purpose
Opcode	1-2 bytes	Identifies the operation, Such as MOV,SUB,ADD,etc.

Mod r/m	0-1 byte	Specifies register operands and memory addressing
SIB	0-1 byte	Specifies scaled-index memory addressing
Displacement	0,1 or 4 bytes	Specifies an offset added to a memory address
Immediate	0,1 or 4 bytes	Contains a constant value used by the instruction

Encoding Lookup Table :

32 bit Registers	Decimal	Binary Code
EAX	0	000
ECX	1	001
EDX	2	010
EBX	3	011
ESP	4	100
EBP	5	101
ESI	6	110
EDI	7	111

Opcode Table :

Index	Mnemonics	Opcode	No. of Operands	Op1	Op2	Mod r/m	3-5 bits	+ rd
A	ADD	01 /r	2	r/m	r	Yes	r	No
	ADD	03 /r	2	r	r/m	Yes	r	No
	ADD	81/0 id	2	r/m	imm	Yes	000	No
	ADD	05 id	2	EAX	imm	No	-1	No
	AND	21 /r	2	r/m	r	Yes	r	No
	AND	23 /r	2	r	r/m	Yes	r	No
	AND	25 id	2	EAX	imm	No	-1	No
	AND	81 /4 id	2	r/m	imm	Yes	100	No
C	CMP	39 /r	2	r/m	r	Yes	r	No
	CMP	3B /r	2	r	r/m	Yes	r	No
	CMP	3D id	2	EAX	imm	No	-1	No
	CMP	81 /7 id	2	r/m	imm	Yes	111	No
D	DEC	r/m	1	r/m	-	Yes	001	No
	DEC	48 + rd	1	r	-	No	-1	Yes
	DIV	F7 /6	1	r/m	-	Yes	110	No
I	INC	FF /0	1	r/m	-	Yes	000	No
	INC	40 + rd	1	r	-	No	-1	Yes
M	MOV	89 /r	2	r/m	r	Yes	r	No
	MOV	8B /r	2	r	r/m	Yes	r	No

	MOV	C7 /0 id	2	r/m	imm	Yes	000	No
	MOV	B8 + rd id	2	r	imm	No	-1	Yes
	MUL	F7 /4	1	r/m	-	Yes	100	No
O	OR	09 /r	2	r/m	r	Yes	r	No
	OR	0B /r	2	r	r/m	Yes	r	No
	OR	0D id	2	EAX	imm	No	-1	No
	OR	81 /1 id	2	r/m	imm	Yes	001	No
S	SUB	29 /r	2	r/m	r	Yes	r	No
	SUB	2B /r	2	r	r/m	Yes	r	No
	SUB	2D id	2	EAX	imm	No	-1	No
	SUB	81 /5 id	2	r/m	imm	Yes	101	No
X	XOR	31 /r	2	r/m	r	Yes	r	No
	XOR	33 /r	2	r	r/m	Yes	r	No
	XOR	35 id	2	EAX	imm	No	-1	No
	XOR	81 /6 id	2	r/m	imm	Yes	110	No

Instruction Encoding Linked-List Structure :

A

│

 |— ADD

│ │

│ ├— 01 /r → r/m32, r32

│ ├— 03 /r → r32, r/m32

│ ├— 05 id → EAX, imm32

│ └— 81 /0 id → r/m32, imm32

│

└— AND

 │

 ├— 21 /r → r/m32, r32

 ├— 23 /r → r32, r/m32

 ├— 25 id → EAX, imm32

└── 81 /4 id → r/m32, imm32

C

│

└── CMP

  │

  ├── 39 /r → r/m32, r32

  ├── 3B /r → r32, r/m32

  ├── 3D id → EAX, imm32

  └── 81 /7 id → r/m32, imm32

D

│

├── DEC

│ │

│ ├── FF /1 → r/m32

│ └── 48 + rd → r32

│

└── DIV

  │

  └── F7 /6 → r/m32

I

│

└── INC

  │

  ├── FF /0 → r/m32

  └── 40 + rd → r32

M

```
│
├── MOV
│   │
│   ├── 89 /r      → r/m32, r32
│   ├── 8B /r      → r32, r/m32
│   ├── C7 /0 id   → r/m32, imm32
│   └── B8 + rd id → r32, imm32
│
└── MUL
    │
    └── F7 /4      → r/m32
```

O

```
│
└── OR
    │
    ├── 09 /r      → r/m32, r32
    ├── 0B /r      → r32, r/m32
    ├── 0D id      → EAX, imm32
    └── 81 /1 id   → r/m32, imm32
```

S

```
│
└── SUB
    │
    ├── 29 /r      → r/m32, r32
    ├── 2B /r      → r32, r/m32
```

├— 2D id → EAX, imm32

└— 81 /5 id → r/m32, imm32

X

│

└— XOR

│

├— 31 /r → r/m32, r32

├— 33 /r → r32, r/m32

├— 35 id → EAX, imm32

└— 81 /6 id → r/m32, imm32

Github Links :

https://github.com/vaishnavishinde1722/SYSTEM

https://github.com/namita-ghodke13

https://github.com/shital0101

https://github.com/vaishnavishinde1722

https://github.com/kaveripatare218